

Deployment & System Architecture Guide

DOCKER, UVICORN & CLOUDFLARE TUNNEL PRODUCTION SETUP

1. System Requirements & Hardware

- **Operating System:** Linux (Ubuntu 20.04+) or Windows 11
- **GPU Acceleration:** NVIDIA GPU with CUDA 11.8/12.1 (Tested on RTX 3050 6GB Laptop GPU)
- **Python Version:** Python 3.10+
- **Containerization:** Docker 24.0+ & Docker Compose v2

2. Method 1: Docker Compose (Recommended)

Launch the containerized API stack with a single command:

```
git clone https://github.com/Aadithya2222/Logistics-Object-Detection-Model.git cd Logistics-Object-Detection-Model docker compose up --build
```

The API service compiles weights, mounts volume paths, and exposes http://localhost:8000.

3. Method 2: Local Virtual Environment

```
python -m venv .venv .\.venv\Scripts\activate pip install -r requirements.txt uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

4. Method 3: Cloudflare HTTPS Tunnel Setup

To expose the local API to the public internet securely with SSL:

```
.\cloudflared.exe tunnel --url http://localhost:8000
```

Output tunnel URL: https://prices-debug-match-twist.trycloudflare.com.

5. Environment Variables (.env)

```
ROBOFLOW_API_KEY=5YSIVQsasg8jgmgPq2WV PORT=8000 MODEL_PATH=weights/best.pt
```